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**Mapping Emergency Demand:
The Deprivation, Demographic, Disease
Burden and Department Patterns Behind
Frequent A&E Attendance in Scotland**

by

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Executive Summary

A small group of frequent attenders, people with three or more A&E visits a year, accounts for a disproportionate share of demand on Scotland's emergency departments. This report models twelve months of Public Health Scotland attendance data across all 14 Health Boards to show who this group is and where pressure concentrates.

Age is the clearest driver. Attendance is lowest among 40–64 year-olds and highest among the over-75s, who attend 74% more often than under-18s. Deprivation doesn't follow the usual pattern. Quintile 3, the middle deprivation group, attends 22% more often than the most deprived group. We treat this as provisional, given a known limitation in the population data. Department type matters more than any demographic factor measured: minor injury units show attendance rates 89% lower than core emergency departments, reflecting patient case mix rather than genuine risk. Three Health Boards stand out once these factors are accounted for: Dumfries and Galloway attend around 1.4 times the expected rate, while Tayside and Forth Valley attend roughly half and two-thirds of the expected rate respectively. These findings were checked against alternative model specifications and remained stable throughout.

Three findings seem most worth acting on. Tayside, Forth Valley, and Dumfries and Galloway stand out as the clearest candidates for local service-access review, since their attendance patterns cannot be explained by the factors measured here. Department type appears better suited to a structural planning variable within Whole System Modelling than a background factor, given how much variation it explains. The deprivation result is best treated as provisional rather than a basis for resourcing decisions, pending deprivation-specific population estimates.

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- I acknowledge the use of GPT-5 to assist in debugging R code error used for data preprocessing, and the generation of figures.
- I acknowledge the use of Grammarly and GPT-5 to check grammar and overall writing quality throughout the final report, ensuring clarity and professionalism.

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1 Introduction

Public Health Scotland (PHS) leads public health work in Scotland, with a focus on prevention and the wider factors that shape health. Accident and Emergency (A&E) is often a patient's first point of contact in a crisis, so meeting the Scottish Government's four-hour A&E target is a key priority [1]. Under that target, 95% of patients must be seen, admitted, discharged, or transferred within four hours, and individual Health Boards aim higher still, at 98%. To support this, PHS uses Whole System Modelling (WSM) to give government and Health Boards evidence for planning.

WSM modelling has focused on operational factors, such as bed availability, since Health Boards control these directly. But evidence suggests these factors explain less of the picture over time. In England, operational variables accounted for only 19% of the variation in four-hour target performance in 2012, falling to 12% by 2016 [2]. This decline points to a growing role for factors outside hospital operations, including population characteristics and access to healthcare, and motivates a closer look at who is driving A&E demand rather than only how hospitals manage it.

A&E demand is not spread evenly across patients. A small group of frequent attenders, typically defined as those with three or more attendances, accounts for a disproportionate share of visits, concentrating pressure on capacity rather than spreading it [3]. Understanding who becomes a frequent attender, and why, is a more targeted question for prevention-focused resourcing than modelling overall attendance volume. Interventions aimed at this group, such as case management, community chronic disease support, or improved primary care access, can plausibly reduce avoidable repeat visits without requiring system-wide capacity expansion.

Demographic characteristics and deprivation are established predictors of A&E attendance more broadly. Poorer access to primary care is associated with higher A&E attendance [4] and with greater reliance on out-of-hours services [5]. Deprivation remains among the strongest predictors of attendance even after accounting for demographic characteristics and disease prevalence, though substantial variation across geographic areas remains unexplained [6]. Whether these gradients hold for Scottish Health Boards specifically remains an open question.

These demographic and deprivation gradients may not fully explain who becomes a frequent attender. Frequent attendance has also been linked to mental health problems, self-harm, alcohol use, and respiratory or digestive illness, whereas cardiac disease and cancer appear protective, likely reflecting effective management through outpatient and community care [3]. This suggests repeat attendance reflects how well specific conditions are managed outside emergency care rather than chronic illness alone. Disease burden therefore looks like a complex but plausible predictor at Health Board level. Whether these gradients, together with disease burden, hold consistently for frequent attendance across Scottish Health Boards is the empirical question this report addresses.

This study quantifies how demographic characteristics, deprivation, and Health Board-level disease burden jointly shape the risk of frequent A&E attendance in Scotland, accounting for variation across Health Boards through a multilevel modelling framework. The aim is to give Public Health Scotland statistically grounded evidence to support prevention-focused resourcing and the continued development of the Whole System Modelling A&E programme.

2 Data and Exploration

2.1 Data Sources

For this study, we use a secondary dataset of A&E attendance counts from Public Health Scotland’s Multiple Attendances Demographics dataset, part of its Monthly A&E Activity and Waiting Times series. “Monthly” here refers to how often the dataset is refreshed, not the window over which attendances are counted. Each release reports counts over a rolling twelve-month period, and the data used in this study covers the twelve months ending April 2026, across all 14 Health Boards in Scotland. The data is aggregated to the level of Health Board, age band, sex, deprivation quintile, and department type: each such combination, referred to throughout this report as a stratum, gives 1,576 rows in total, recording, for each one, how many individuals attended exactly once, twice, three, four, or five or more times. Because a person’s age group or deprivation category can change within a rolling twelve-month window, the same individual may be counted under more than one such category. Totals here should therefore be read as a count of attendance records, rather than a strict count of unique people. After excluding strata with a missing age, sex, or deprivation value (Section 2.2.2), our sample comprises 1,094,829 individuals, of whom 86,353 (7.9%) were frequent attenders, defined as those with three or more attendances within the period.

Disease prevalence at Health Board level, sourced from Public Health Scotland’s General Practice disease prevalence dataset, is used to capture how much chronic illness might explain frequent attendance at each Health Board. This records how common ten long-term conditions, including heart failure, dementia, and depression, are within each Health Board, using the most recent year available, in 2025.

2.2 Data Preprocessing

2.2.1 Deriving Frequent Attendance

We define a frequent attender as anyone with three or more A&E attendances within the twelve-month period, in line with common practice in the frequent-attender literature [3], though this sits below Public Health Scotland’s own clinical monitoring definition of ten or more attendances within a year, or five or more within any three-month period [7]. That three-month criterion needs the exact date of each attendance, and this data simply doesn’t record it. For each stratum, it gives only how many individuals attended once, twice, three, four, or five or more times over the full year. An annual-count threshold is therefore the only one this data can support, and it comes with the added benefit of capturing a broader, earlier-stage group of repeat attenders than PHS’s stricter standard.

Let $n_{k,i}$ denote the number of individuals in stratum i attending exactly k times ($k = 1, 2, 3, 4$), and let $n_{5+,i}$ denote those attending five or more times. The total number of individuals and the number of frequent attenders in stratum i then follow directly:

$$\text{Total}_i = n_{1,i} + n_{2,i} + n_{3,i} + n_{4,i} + n_{5+,i}, \quad (2.1)$$

$$\text{Frequent}_i = n_{3,i} + n_{4,i} + n_{5+,i}. \quad (2.2)$$

2.2.2 Data cleansing

Three exclusions were applied. Records with missing age, sex, or deprivation were excluded. Missing age and sex reflected placeholder rows, so removing them raises no concerns, but missing deprivation depends on having a fixed Scottish postcode, occurring specifically for people who are homeless, tourists, and short-term visitors. As Figure 1 shows, this asymmetry is substantial, with age and sex each missing for 1.4% of records against 17.1% for deprivation. Drawing on Rubin [8]’s framework, we treat this as Missing Not at Random (MNAR) rather than a random gap in the data. The reason a value goes missing, not having a fixed address, is plausibly tied to someone’s underlying health risk [9], not to anything the dataset already captures elsewhere.

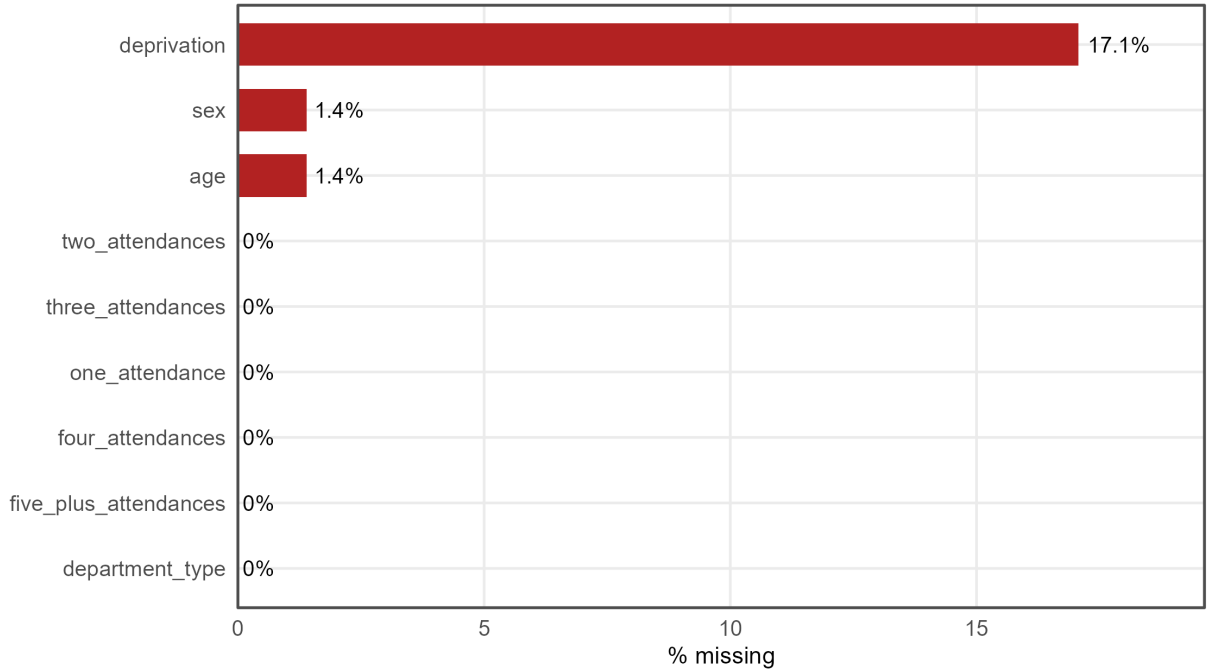


Figure 1: Percentage of missing values by variable in the raw multiple-attendance extract, before any exclusions were applied.

This excluded group is not one single population. It likely includes homeless individuals, whose reliance on unscheduled care, and tourists unlikely to accumulate three or more attendances during a brief stay. Correcting formally for an MNAR mechanism would require external data on the missingness process itself, which isn't available here. We therefore proceed under a Missing At Random (MAR) assumption and analyse complete cases directly, treating this simplification as a limitation rather than something we've resolved statistically. We didn't attempt imputation, since assigning a deprivation value to people without a permanent address would fabricate a profile for them.

2.3 Exploratory Data Analysis

2.3.1 Identifying the Distribution of Frequent Attendance

Figure 2 illustrate how frequent-attender counts are spread across the 1,307 strata that remain after cleansing. The picture is heavily right-skewed. Most strata contain only a handful of frequent attenders, giving the tall spike near zero, while a smaller number stretch the distribution out past 500, and occasionally beyond 1,000. The average count across strata is 66.1, but the spread around that average is far wider than the average itself, with a standard deviation of 130.5.

This gap between the average and its spread is not incidental. It points directly to the choice of statistical model described in Section 3.1. A Poisson distribution, the natural starting point for count data, assumes the variance equals the mean. Here the variance is 17,024.9, around 258 times the mean, so that assumption clearly doesn't hold. A Poisson model would understate how much strata genuinely differ from one another, producing confidence intervals too narrow to be trusted. This is why we used a negative binomial model instead, which allows the variance to exceed the mean through a separate dispersion parameter.

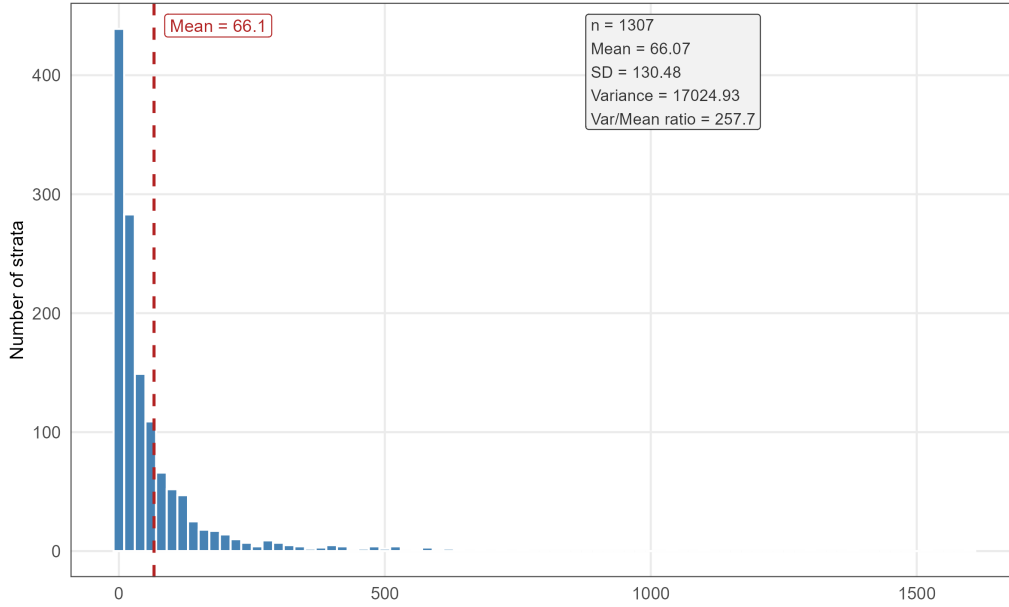


Figure 2: Distribution of frequent-attender counts per stratum ($n = 1,307$).

2.3.2 Demographic and deprivation patterns in frequent attendance

Before fitting the model, we first look at how frequent attendance actually varies across demographic groups in the raw data, since these patterns should motivate, and later help sense-check, what the model finds. Figure 3 breaks the proportion of frequent attenders down by deprivation quintile, age group, and sex. One pattern holds in every single panel: attendance falls steadily from the most to the least deprived quintile, with no exceptions across any age group or sex, consistent with deprivation’s established role as a strong, general predictor of A&E use [6]. This consistency matters. It suggests deprivation’s effect on frequent attendance does not depend on a person’s age or sex, an encouraging sign for a model that treats these three predictors as separate, additive effects rather than needing to model how they interact.

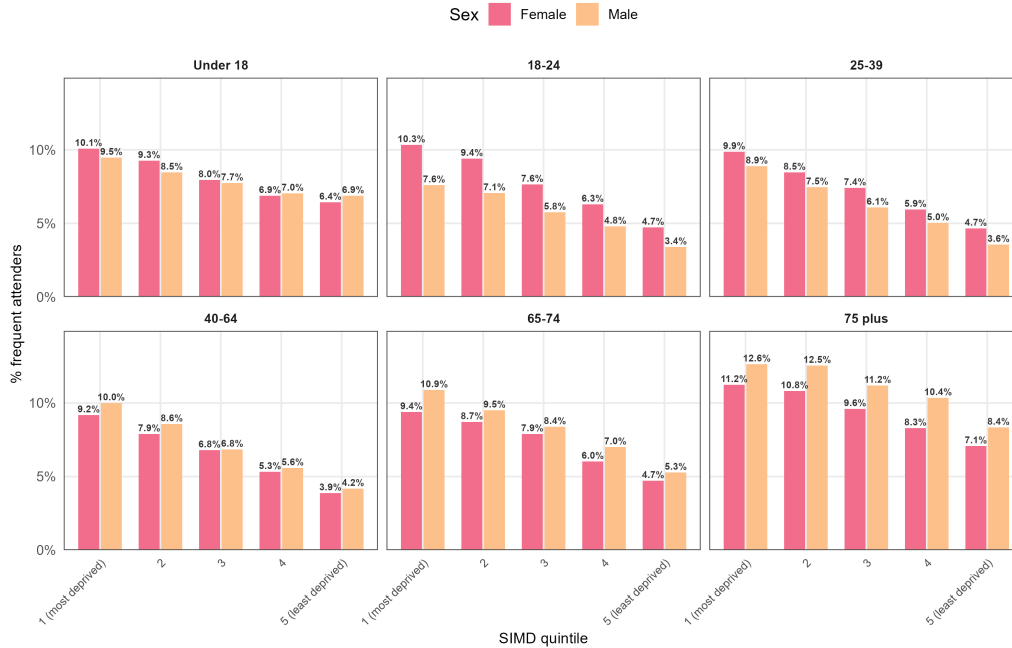


Figure 3: Proportion of individuals with three or more A&E attendances, by SIMD deprivation quintile, age group, and sex

Age and sex tell a different story, though, and one worth flagging on its own terms. Among children and younger adults, up to around age 39, women attend more frequently than men at every deprivation level. Past midlife, this pattern reverses. From age 40 onward, men consistently show higher rates than women, most visibly among the over-75s, where male attendance reaches 12.6% in the most deprived quintile against 11.2% for women. This crossover echoes wider evidence that age and sex both shape emergency department use in shifting, non-uniform ways [10]. Unlike deprivation’s steady, one-directional effect, sex’s relationship with frequent attendance clearly shifts across the life course.

2.3.3 Disease prevalence and frequent attendance

In this section, we check how the Board-level proportion of frequent attenders relates to each of the ten candidate disease-prevalence rates, and how those rates relate to one another, using Pearson correlation. Pearson’s coefficient is the natural choice here, since every variable involved is a continuous rate rather than a category. Figure 4 explains the results. Only a handful of conditions show any meaningful link to frequent attendance. Heart failure ($r = 0.40$) and depression ($r = -0.35$) stand out, positive and negative respectively, followed some way behind by palliative care ($r = 0.29$) and coronary heart disease ($r = 0.26$). Most others, including diabetes, epilepsy, and mental health, sit close to zero. With only 14 Health Boards, these correlations are indicative rather than confirmatory, but they offer an early indication of which conditions are worth carrying forward.

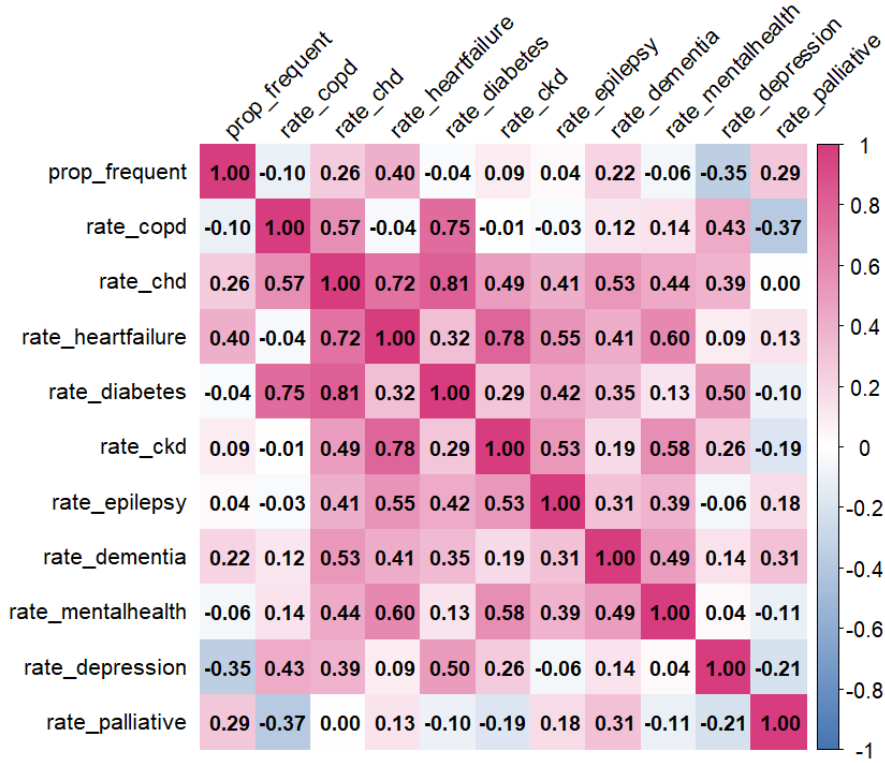


Figure 4: Pearson correlation matrix between Board-level frequent attender proportion and candidate disease prevalence rates

A second, more consequential pattern is that several candidate predictors are highly correlated with one another, not just with the outcome. Coronary heart disease correlates above 0.7 with both diabetes ($r = 0.81$) and heart failure ($r = 0.72$), and heart failure correlates 0.78 with chronic kidney disease. Correlations at this level are a recognised warning sign for multicollinearity, generally treated as a concern once $|r|$ exceeds around 0.7 [23]. Entering all ten rates into a single model would inflate the variance of their coefficients, making it difficult to attribute any effect to one specific condition rather than to whichever correlated conditions travel alongside it. This is why, not all ten variables were carried forward into the final model. This correlation structure, together with variance inflation

checks, was used to narrow the list to a smaller, less redundant set.

2.3.4 Population adjustment

We account for population information at Health Board, age group, and sex level, sourced from National Records of Scotland’s mid-year 2025 population estimates, the most recent available. Health Boards vary hugely in population size (Figure 5). Without adjusting for this, a large board such as NHS Greater Glasgow and Clyde would almost always show more frequent attenders than a small board such as NHS Orkney, simply because far more people live there, not because people attend A&E more often. This relies on an assumption worth stating explicitly, that the offset itself is independent of the true underlying rate. Conn and Ferguson [11] show that predictions can become biased when the offset is itself linked to the process being estimated. This typically happens when the offset comes from a measurement process, such as detection probability in certain surveys, that is correlated with the outcome. Population size is different in kind, since it is set by demography rather than by any process connected to A&E attendance, so this assumption is considered reasonable here, though noted as a modelling choice rather than guaranteed.

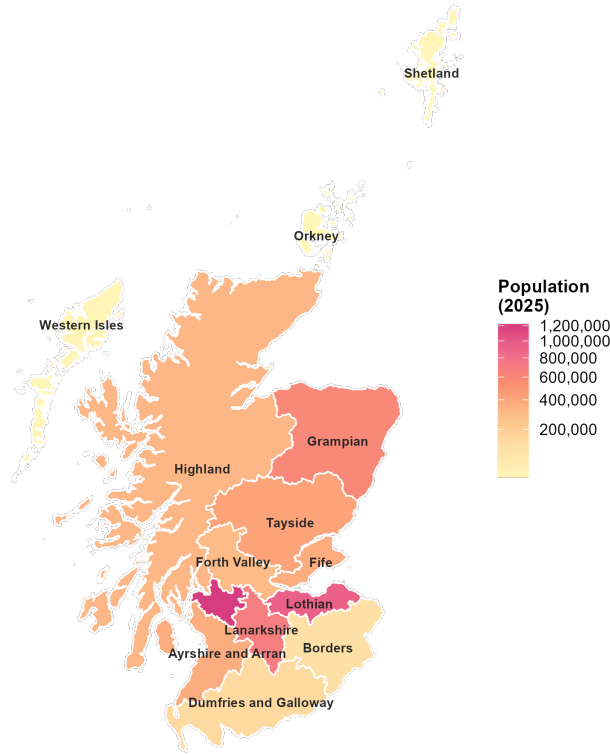


Figure 5: Population distribution across NHS Scotland Health Boards, mid-year 2025 estimates.

3 Method

3.1 Model construction

The number of frequent attenders in each stratum was modelled as a count, with the log of the underlying population included as an offset, so the model estimates a *rate* per head of population rather than reflecting where more people happen to live. Health outcome counts of this kind rarely satisfy the assumptions of ordinary linear regression, since they are non-negative, discrete, and often right-skewed, properties better captured directly by count distributions such as the Poisson and negative binomial than by transforming the outcome and fitting a standard linear model [12].

As shown in Section 2.3.1, a preliminary variance-to-mean check confirmed substantial overdispersion in the frequent-attender counts, so a negative binomial distribution was used in place of a standard Poisson model [13]. A random intercept for Health Board was included, since attendances within the same board likely share unmeasured local influences and are not fully independent [14]. Age, sex, deprivation, and department type were entered as categorical predictors against a reference level, while only disease-prevalence rates entered as continuous. Age has six bands, with Under 18 as the reference. Deprivation has five SIMD quintiles, with quintile 1 (most deprived) as the reference. Sex has two categories, Female and Male, with Female as the reference. Department type has two categories present in this extract, Type 1 and Type 3, with Type 1 as the reference. Because sex and department type each have only one non-reference level, they contribute a single indicator term rather than a sum.

Population estimates from National Records of Scotland are available by Health Board, age, and sex, but not further broken down by deprivation or department type. The population offset therefore varies at a coarser level than the full stratum i . Let $g(i)$ denote the Health Board, age, and sex group to which stratum i belongs, so that every stratum sharing the same board, age band, and sex, but differing in deprivation or department type, shares the same population value $\text{Population}_{g(i)}$. The implications of this for interpreting the deprivation and department-type coefficients are discussed in the Limitations section. The resulting negative binomial Generalised Linear Mixed Model (GLMM), fitted using `glmmTMB` [15], took the form, for stratum i nested within Health Board $b(i)$:

Formulas:

$$Y_i \mid u_{b(i)} \sim \text{NegBin}(\mu_i, \theta), \quad (3.1)$$

$$\begin{aligned} \log(\mu_i) = & \log(\text{Population}_{g(i)}) + \beta_0 + \sum_{a=2}^6 \beta_{\text{age},a} \text{I}(\text{Age}_i = a) + \beta_{\text{sex}} \text{I}(\text{Sex}_i = \text{Male}) \\ & + \sum_{d=2}^5 \beta_{\text{dep},d} \text{I}(\text{Deprivation}_i = d) + \beta_{\text{dept}} \text{I}(\text{DeptType}_i = \text{Type 3}) \\ & + \beta_{\text{disease}}^\top \mathbf{Disease}_{b(i)} + u_{b(i)}, \end{aligned} \quad (3.2)$$

$$u_b \sim \mathcal{N}(0, \sigma_u^2), \quad b = 1, \dots, 14. \quad (3.3)$$

Notes:

Y_i = observed frequent-attender count in stratum i

μ_i = model's expected count for stratum i

θ = dispersion parameter, allowing variance to exceed the mean (Section 2.3.1)

$\beta_{\text{age},a}, \beta_{\text{sex}}, \beta_{\text{dep},d}, \beta_{\text{dept}}$ = effect of that category vs. its reference level, not a slope

$\mathbf{Disease}_{b(i)}$ = continuous disease-prevalence rates for Health Board $b(i)$

u_b = HBT random intercept

Exponentiating any coefficient gives a rate ratio. There is no separate error term in the model. All of the randomness comes from the negative binomial distribution itself, plus u_b .

Indexes:

- i = a stratum, defined by Health Board, age, sex, deprivation, and department type
 $b(i)$ = the Health Board to which stratum i belongs
 $g(i)$ = the Health Board, age, and sex group to which stratum i belongs (see above)

3.2 Model Selection

With six candidate predictors available, letting software search through every possible combination and retaining whichever fits best is tempting but hard to justify: this kind of blind search tends to overstate significance and distort coefficient estimates, precisely because the same data are used to both choose and then interpret the model [16]. Instead, predictors were added in four blocks. Age and sex entered first, followed by deprivation, department type, and finally the two disease-prevalence rates carried forward. Each block was tested against the model above it using a likelihood ratio test, a formal statistical comparison of whether adding that block genuinely improves the model, and using AIC (Akaike Information Criterion), a single score that summarises how well a model balances fit against complexity, with a lower score indicating a better trade-off [14]. Every group of variables therefore had to earn its place rather than being included on assumption alone.

Table 1 summarises the result. Every block reached significance, with department type contributing by far the largest single improvement in fit ($\chi^2 = 1008.81$, $df = 1$, $p < .001$) and the biggest drop in AIC, consistent with the size of its coefficient in the final results. The full model, M4, is therefore taken forward as the final specification used throughout the remainder of this report. Its assumptions were subsequently checked using simulation-based residual diagnostics, discussed in the Appendix.

Table 1: Model comparison for selecting final predictors.

Model	Predictors	Df	AIC	χ^2	χ^2 df	p
M0	Offset and (1 hbt) only	3	12301	—	—	—
M1	M0 + age, sex	9	12231	82.23	6	<.001***
M2	M1 + deprivation	13	12134	104.41	4	<.001***
M3	M2 + department type	14	11128	1008.81	1	<.001***
M4	M3 + heart failure, depression rate	16	11124	7.70	2	.021*

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Each χ^2 test compares the null hypothesis that the added block of predictors has no effect (all added coefficients equal zero) against the alternative that at least one does, so a significant result means that block improves the model's fit over the row above it, not that every individual predictor within that block is itself significant.

3.3 Sensitivity Analysis for Final Model : Zero Inflation and Parameter Stability

To check whether the zero-underprediction threatens our conclusions, we compared three specifications, the Poisson model, the primary negative binomial model (M4), and a zero-inflated extension with department type in the zero-inflation component. Comparing nested count models this way on fit and coefficient stability follows standard practice [14].

Once we compare the models, the overdispersion is obvious. Poisson's AIC (34,263) sits far above both M4 (11,124) and the Zero-Inflated Negative Binomial (ZINB) extension (11,107), and it predicts only 37 zero-count strata against the 116 actually observed, compared with 60 under M4 and 82 under the ZINB extension (Figure 6).

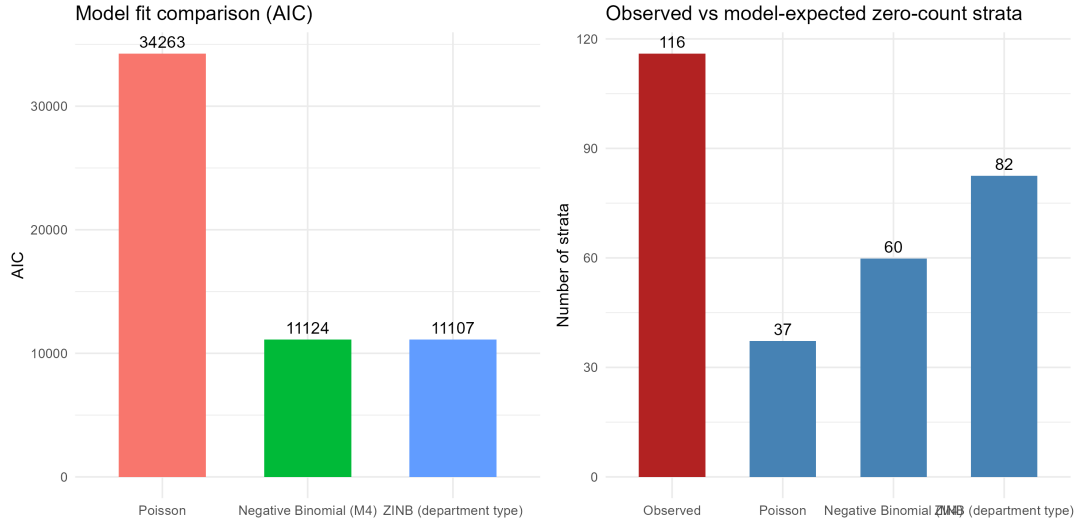


Figure 6: Model fit and zero-count comparison across Poisson, negative binomial (M4), and the department-type ZINB extension

Comparing the fixed-effect coefficients directly makes the same point from another angle. Poisson’s confidence intervals are noticeably tighter than M4’s or the ZINB extension’s, the usual sign of ignored overdispersion [13], and a few point estimates, deprivation quintile 5 in particular, sit clearly apart from the other two models. M4 and the ZINB extension agree closely on almost every predictor. Heart failure prevalence is the only coefficient that shifts meaningfully between them (Figure 7).

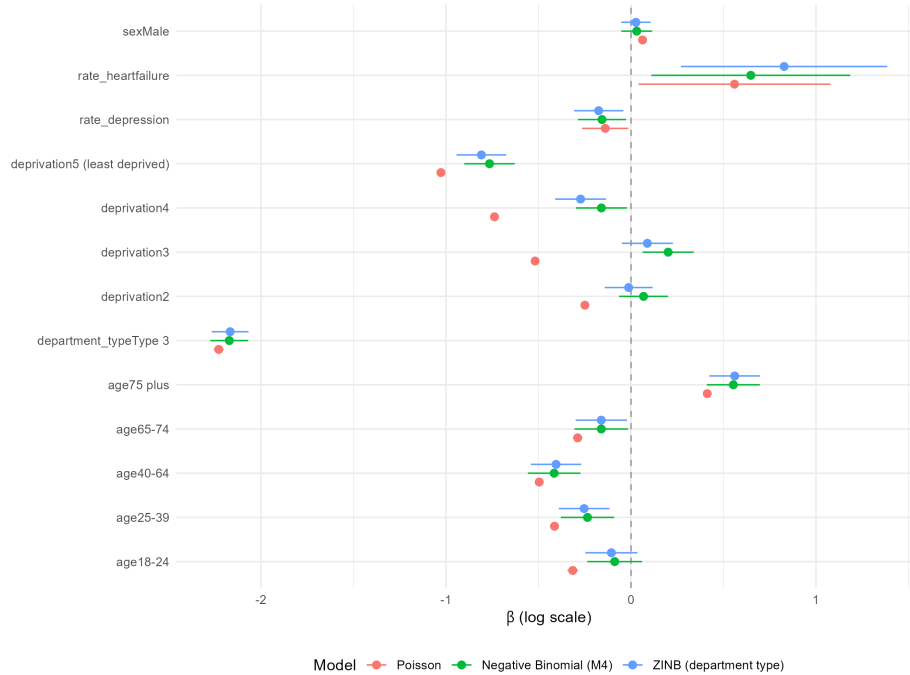


Figure 7: Fixed-effect coefficients (log scale) with 95% confidence intervals, compared across Poisson, negative binomial (M4), and the department-type ZINB extension.

The model slightly underestimates how many age/deprivation/board groups have no frequent attenders at all. When we accounted for department type separately, most of this gap closed, confirming that department type is the main reason some groups show zero frequent attenders. A small gap remains, likely from very small or unusual combinations of age, deprivation, and board that are too specific for the model to capture individually. This is minor enough that it doesn’t change any of the findings reported in this report (see Appendix A.2).

4 Result and Discussion

From the final model we obtain, for each demographic, deprivation, department, and disease-prevalence category, an estimated effect on the rate of frequent attendance, expressed relative to a reference group, such as each age band compared with under-18s. Exponentiating these estimates converts them into rate ratios, showing how much more, or less, often that group attends than the reference group. Table 2 reports these rate ratios along with their confidence intervals (CI) and significance. The subsections that follow work through each block of predictors in turn, covering demographic effects, deprivation, department type, disease prevalence, and the differences between Health Boards Treatment.

Table 2: Fitted coefficients and rate ratios from the final negative binomial GLMM ($n = 1,307$ strata, 14 Health Boards, random intercept variance $\sigma_u^2 = 0.096$).

Predictor	β	Rate ratio	95% CI of Rate Ratio	p
Age (ref: Under 18)				
18–24	−0.088	0.92	0.79–1.06	.242
25–39	−0.235***	0.79	0.68–0.91	.001
40–64	−0.415***	0.66	0.57–0.76	<.001
65–74	−0.161**	0.85	0.74–0.98	.030
75 plus	0.553***	1.74	1.51–2.01	<.001
Sex (ref: Female)				
Male	0.031	1.03	0.95–1.12	.467
Deprivation (ref: Quintile 1, most deprived)				
Quintile 2	0.068	1.07	0.94–1.22	.315
Quintile 3	0.201**	1.22	1.06–1.40	.005
Quintile 4	−0.160*	0.85	0.74–0.98	.023
Quintile 5 (least deprived)	−0.765***	0.47	0.41–0.53	<.001
Department type (ref: Type 1)				
Type 3	−2.172***	0.11	0.10–0.13	<.001
Health Board disease prevalence				
Heart failure rate	0.648*	1.91	1.12–3.27	.018
Depression rate	−0.157*	0.86	0.75–0.97	.018

significant at *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

4.1 Demographic Effects

Age was the strongest demographic predictor in the model. Frequent attendance was lowest among the 40–64 group, 34% below the under-18 reference (95% CI: 24–43% lower), and highest among the over-75s, who attended 74% more often (95% CI: 51–101% higher), both statistically significant. The 18–24 group was the only age band not significantly different from the under-18 reference (8% lower, 95% CI: 21% lower to 6% higher, $p = .242$). This pattern is consistent with Reid et al. [3], who found frequent attendance concentrated at the extremes of the life course, and with the trimodal pattern Marshall et al. [17] identified in a national English cohort, with peaks among young adults, those in mid-life, and older adults.

Sex, on its own, was not a significant predictor (rate ratio 1.03, $p = .467$). Since age and sex were entered as separate, additive terms rather than an interaction, this null result may reflect an effect that varies across the life course rather than a genuinely absent one [10]. Testing an age by sex interaction would clarify whether this is the case.

4.2 Deprivation

Quintile 3 recorded the highest frequent-attendance rate of any deprivation group, 22% above the most deprived reference (95% CI: 6–40% higher, $p = .005$), while quintile 5, the least deprived group, fell to

53% below the reference rate (95% CI: 47–59% lower, $p < .001$), both statistically significant. Quintile 4 also reached significance, 15% lower (95% CI: 2–26% lower, $p = .023$), but quintile 2 did not differ significantly from the most deprived reference (7% higher, 95% CI: 6% lower to 22% higher, $p = .315$). Quintiles 2 and 4 therefore did not sit neatly in between the two extremes, giving deprivation a non-monotonic effect rather than the steady, one-directional gradient more commonly reported [6]. LaCalle and Rabin [18] caution against exactly this kind of assumption, arguing that common beliefs about who frequently attends are not always well supported once the data are examined closely, and the result here fits that caution.

Two features of the model may be driving this. The population offset is coarser than ideal, and department type isn’t evenly spread across deprivation groups (see Section 6). We’d treat this result as provisional rather than conclusive.

4.3 Department Type

Type 3 departments were associated with an 89% lower rate of frequent attendance than Type 1 (95% CI: 87–90% lower, $p < .001$), a highly significant and by far the largest effect of any predictor in the model. A&E sites in Scotland are classified into three types. Type 1 sites are core emergency departments, open 24 hours a day and consultant led, with full resuscitation facilities for serious and life-threatening conditions. Type 2 sites are single specialty emergency departments, for example ophthalmology, dental, psychiatry or obstetrics. Type 3 sites are non-core minor injuries units, usually GP or nurse led, sometimes with limited opening hours [1]. Only Type 1 and Type 3 attendances appear in this extract. The scale of the gap makes sense given how these sites actually function. Type 3 departments handle only minor injuries and illnesses, so the recurring, complex presentations that drive frequent attendance are concentrated almost entirely at Type 1 sites. This lines up with Rudge et al. [19]’s finding that a nearby minor injuries unit measurably diverts demand away from the main emergency department.

Department type is closely tied to where more unwell patients get seen, so part of the age and deprivation effects reported above may be partly explained by this routing rather than fully independent of it. The nested model comparison in Model Selection (Section 3.2) already points to this. Department type contributed by far the largest single improvement in fit of any block added to the model, underlining how much explanatory weight it carries relative to the demographic and deprivation predictors entered before it.

4.4 Disease prevalence

Heart failure prevalence moved in step with frequent attendance: each one-point rise in a Health Board’s prevalence corresponded to a rate roughly 91% higher (95% CI: 12–227% higher, $p = .018$), a significant but imprecisely estimated effect given the width of that interval. This fits heart failure’s pattern of flare and remission, and its recognised role as a leading cause of unplanned readmission [20]. It sits in some tension with Reid et al. [3], who reported cardiac disease as broadly protective against frequent attendance at the individual level. Heart failure is a narrower, more severe condition within that broader category, though, prone to acute decompensation in a way that better-controlled cardiac disease is not, so the two findings are not necessarily contradictory.

Depression told the opposite story. A one-point rise in a board’s depression prevalence corresponded to a rate around 14% lower (95% CI: 3–25% lower, $p = .018$), also significant, and the reverse of what individual-level studies typically report [21]. Since this association was measured at Health Board level rather than the individual level, boards with higher recorded depression prevalence may also have stronger community mental health provision, of the kind associated with reduced emergency attendance where primary and community access is good [4]. Neither disease-prevalence result should be read as evidence of what happens to any single attender.

4.5 Health Board Treatment Random Effect

After controlling for demography, deprivation, department type, and Health Board disease prevalence, meaningful variation in frequent attendance remains across Scotland’s 14 Health Boards. As Table 3

shows, only three boards have rate ratios statistically distinct.

Table 3: Summary of Health Board random effect rate ratios relative to national expectation, highlighting significant outliers and primary operational takeaways for Public Health Scotland.

Health Board	Rate Ratio	95% CI	Significance	Primary Operational Takeaway
Dumfries & Galloway	1.38	[1.08, 1.76]	Higher Than Expected	Priority candidate for local rural access and unscheduled transport review [19].
Tayside	0.50	[0.39, 0.64]	Lower Than Expected	Audit local primary care, out-of-hours, and community diversion models [5].
Forth Valley	0.64	[0.51, 0.81]	Lower Than Expected	Evaluate ED triage, case-mix routing, and minor injury unit utilization [19].
<i>11 Remaining Boards</i>	<i>0.80–1.34</i>	<i>Spans 1.0</i>	Not Significant	Attendance patterns are adequately explained by demography, deprivation, and department mix.

Dumfries and Galloway attended 1.38 times the predicted rate (95% CI: 1.08–1.76), the only board with significantly elevated demand. Tayside and Forth Valley attended significantly less, at roughly half (Rate Ratio = 0.50, 95% CI: 0.39–0.64) and two-thirds (Rate Ratio = 0.64, 95% CI: 0.51–0.81) of their expected rates. The remaining eleven boards, including Lothian (1.34), Shetland (1.27), and Borders (1.27), have confidence intervals spanning 1.0 (Figure 9), so we cannot say their attendance genuinely differs from what the model predicts once demography and department mix are accounted for. Unexplained variation across health regions is well-documented in emergency care literature [2, 6]. Local service configuration, primary care access, and travel thresholds shape unscheduled care seeking [4]. For Public Health Scotland’s Whole System Modelling programme, this narrows the immediate policy priority list. Rather than reviewing all fourteen boards, resourcing and service-access audits should focus on Dumfries and Galloway, Tayside, and Forth Valley to establish the local mechanisms driving their divergent attendance.

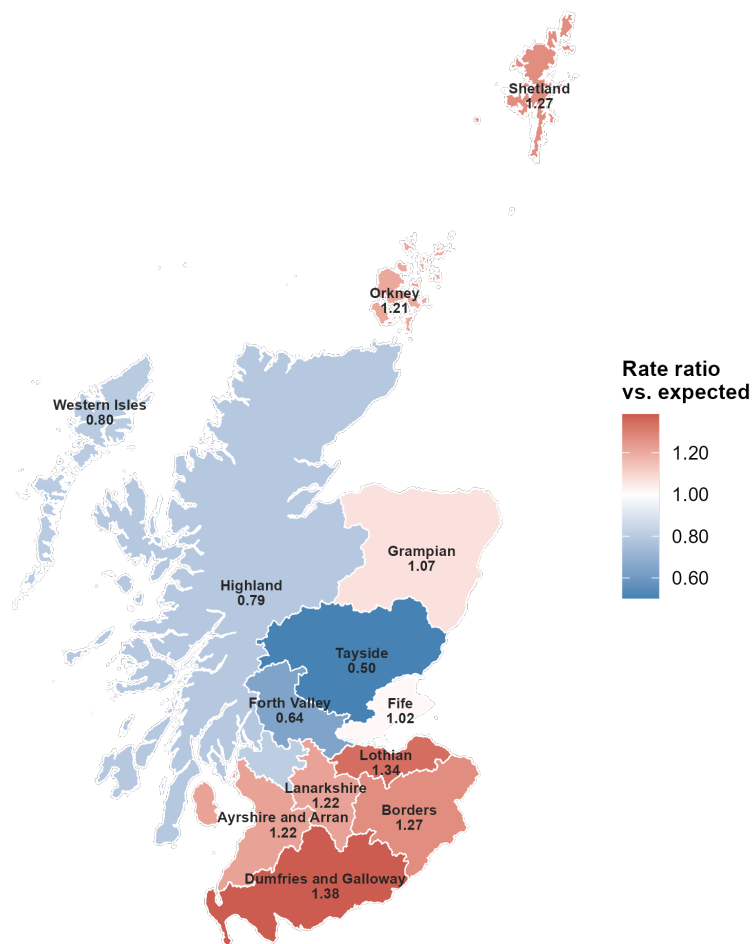


Figure 8: Health Board rate ratios relative to national expectation, after adjusting for age, sex, deprivation, department type, and disease prevalence. Red indicates more frequent attendance than expected, while Blue indicates less

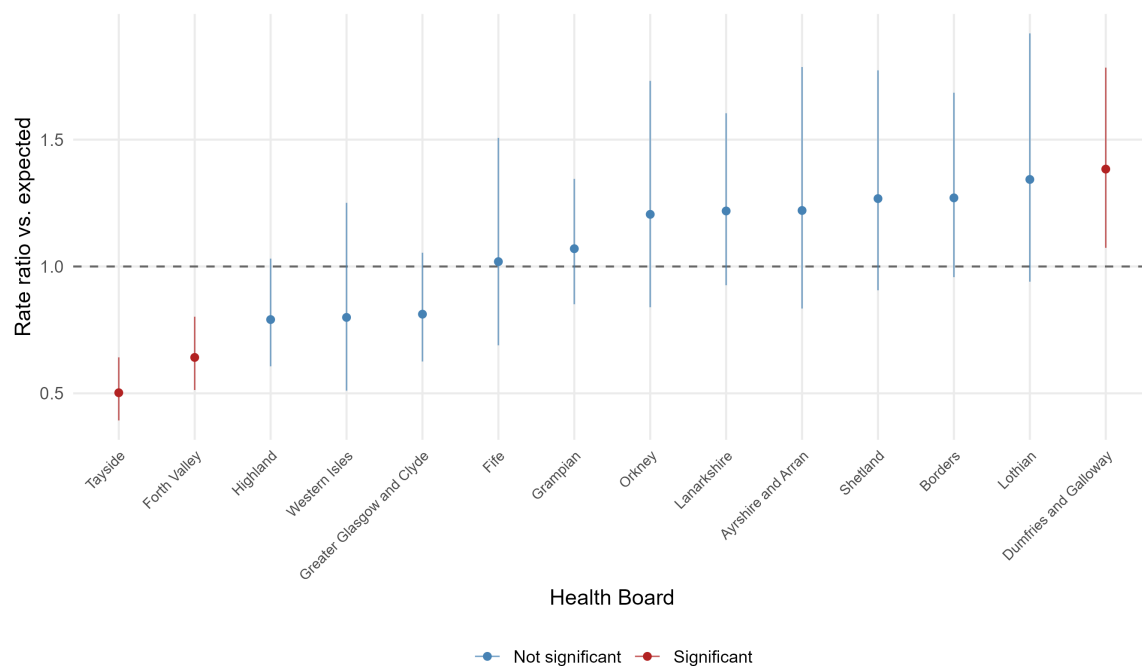


Figure 9: Health Board Rate Ratio in Figure 8 with 95% confidence intervals. The dashed line (value of 1.0) indicates attendance exactly matching that prediction

5 Conclusions

This report set out to quantify how demographic characteristics, deprivation, and Health Board-level disease burden jointly shape the risk of frequent A&E attendance in Scotland, with the aim of giving Public Health Scotland statistically grounded evidence for prevention-focused resourcing and the continued development of the Whole System Modelling programme. Three main conclusions follow from the model.

First, who attends frequently is not evenly spread across the population, and the pattern is more complex than a simple deprivation-driven story. Age remains the clearest and most consistent driver. Attendance dips through working age, then rises sharply among the over-75s (Section 4.1). Deprivation matters too, but its adjusted effect does not rise or fall in a straight line across quintiles. Section 4.2 sets out good reason to treat that result as provisional pending a deprivation-specific population offset, rather than as a finding ready to inform resource allocation on its own [18]. A narrower pattern also emerged within age itself. Men aged 18–24 attend less often than expected once other factors are accounted for, a gap that held on its own even though the wider age-by-sex pattern did not (Appendix A.3). The result is small, but it deserves a closer look before anyone folds it into the general age story.

Second, where a person is seen matters enormously. Department type was the single largest effect in the model, reflecting how differently Type 1 and Type 3 sites are used rather than any demographic risk in itself [19]. This points toward case mix and service design, not just patient characteristics, as central to understanding where frequent-attendance pressure concentrates.

Third, meaningful variation between Health Boards remains even after every measured factor is accounted for, though only three boards, Tayside and Forth Valley on the low side and Dumfries and Galloway on the high side, show a difference precise enough to act on with confidence (Section 4.5). This fits evidence that a growing share of attendance variation comes from factors outside routine operational data [2, 6]. For PHS’s Whole System Modelling work, this narrows the immediate priority list considerably. Rather than treating all fourteen boards as equally worth investigating, the evidence here supports starting with these three, using locally held service and access data to establish what is actually different about them, before extending the same kind of local review elsewhere.

In summary, these results support prevention-focused resourcing that looks beyond deprivation alone. Age-tailored approaches, department-level case mix, and targeted local investigation in the boards where the statistical evidence is strongest all matter more here than a uniform national response would suggest.

6 Limitation and Future Works

The main limitation is the population denominator. National Records of Scotland breaks population estimates down by Health Board, age, and sex, but not by deprivation quintile or department type, so every stratum is divided by the same figure regardless of how deprived it actually is. This likely explains why deprivation departs from the clean upward pattern seen in the raw data. A population breakdown at this finer stratum does exist in principle, cross-tabulated by age, sex, and SIMD decile at Health Board level, but the most recent release available covers only 2021. Re-estimating with a current, fully disaggregated denominator therefore depends on NRS updating that series.

We modelled age and sex additively, then tested that choice. The raw data show a crossover, women attending more below 40, men more from midlife (Appendix A.3). Interacting age and sex improved fit slightly, almost entirely from one group. Men aged 18–24 attended less than the additive model predicted. The other four age bands showed no significant difference. Once deprivation, department type, and disease burden enter, the crossover in Figure 3 narrows to this single gap among young men. We report the additive model as primary, since five extra interaction terms would cost precision elsewhere for a pattern confined to one subgroup. The young-men result still deserves its own flag, since it breaks from the pattern among peers and may warrant closer investigation rather than folding into the general age effect. An age-by-deprivation interaction remains untested, worth exploring given how unevenly deprivation’s effect moved across quintiles.

The two disease-prevalence variables are Health Board averages, not individual patient data, which carries a real risk of ecological fallacy. That probably explains why the depression result runs opposite to individual-level studies [21]. Treat those two coefficients as describing boards, not patients, until we can check them against individual-level data.

We excluded records with missing deprivation data. That pattern of missingness looks consistent with people who lack a fixed address, most plausibly people experiencing homelessness or short-term visitors [9]. Since this group relies more heavily on emergency care, the frequent-attender figures here probably understate the true burden.

Only 14 Health Boards feed the random intercept, so estimates of board-level variance, and which boards differ significantly, carry limited precision. Several years of data, or a smaller unit such as Health and Social Care Partnership, would sharpen that boundary.

The model slightly underestimates how many strata have zero frequent attenders, even though every other diagnostic check held up. A fix built around department type closes most of the gap but not all of it, since minor injury units are the ones most likely to see no frequent attenders at all. The remainder likely comes from small, specific combinations, a particular age group, deprivation level, and Health Board together, too niche for one factor to explain alone. Adding deprivation into that same fix narrows the gap a little further but makes some of its estimates unreliable, so the simpler model is retained as primary, since it stays easy to interpret and none of the conclusions in this report depend on it perfectly predicting zero attenders. This model should therefore not be used to predict which specific strata will have no frequent attenders, for example when deciding where to target outreach services, without further work first.

A further extension would bring in spatial variables not captured here. Studies elsewhere consistently link distance and travel time to the nearest AE, and to alternatives like minor injury units, with attendance rates. This effect also interacts with deprivation, since deprived populations far from a department behave differently to deprived populations close by [19]. A distance or access measure at the A&E level could explain more of the board-level variance, sharpen the picture beyond Health Board alone, and account for some of the residual zero-inflation above. Primary care and out-of-hours access, GP list size, and ambulance response times are worth testing too, all linked to attendance generally, but not yet tested here for frequent attendance specifically.

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A Appendices

A.1 Model Assumption and Diagnostic

After selecting final model in Section 3.2, the next question is whether it is actually valid, that is, whether the negative binomial GLMM meets the assumptions it relies on, rather than simply being the best of the specifications tried. This section checks that using simulation-based residual diagnostics from the DHARMA package, which generates scaled residuals through repeated simulation from the fitted model, giving a distribution-appropriate check that avoids the pitfalls of applying standard linear-model residual diagnostics directly to count data [22].

The first question is whether the residuals follow the distribution the model assumes, the count-data equivalent of checking normally distributed residuals in a linear model. A Kolmogorov-Smirnov test on the scaled residuals showed no evidence of departure from the expected distribution ($p = .227$), so this assumption holds. Next, since the whole rationale for choosing a negative binomial over a Poisson model was to account for overdispersion, that choice itself needs checking rather than being assumed. The dispersion test, comparing the standard deviation of simulated residuals against the fitted model, showed no evidence of remaining over- or underdispersion ($p = .88$), confirming the negative binomial specification adequately captured the excess variance present in the raw counts.

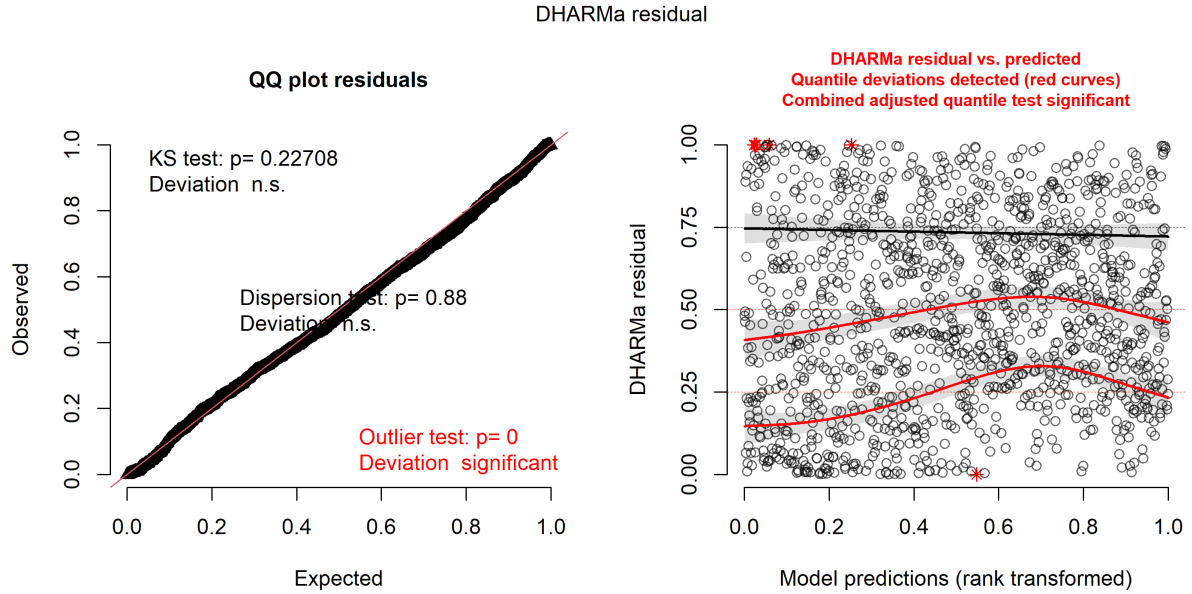


Figure 10: DHARMA simulation-based residual QQ plot and residual vs. predicted plot (1,000 simulations)

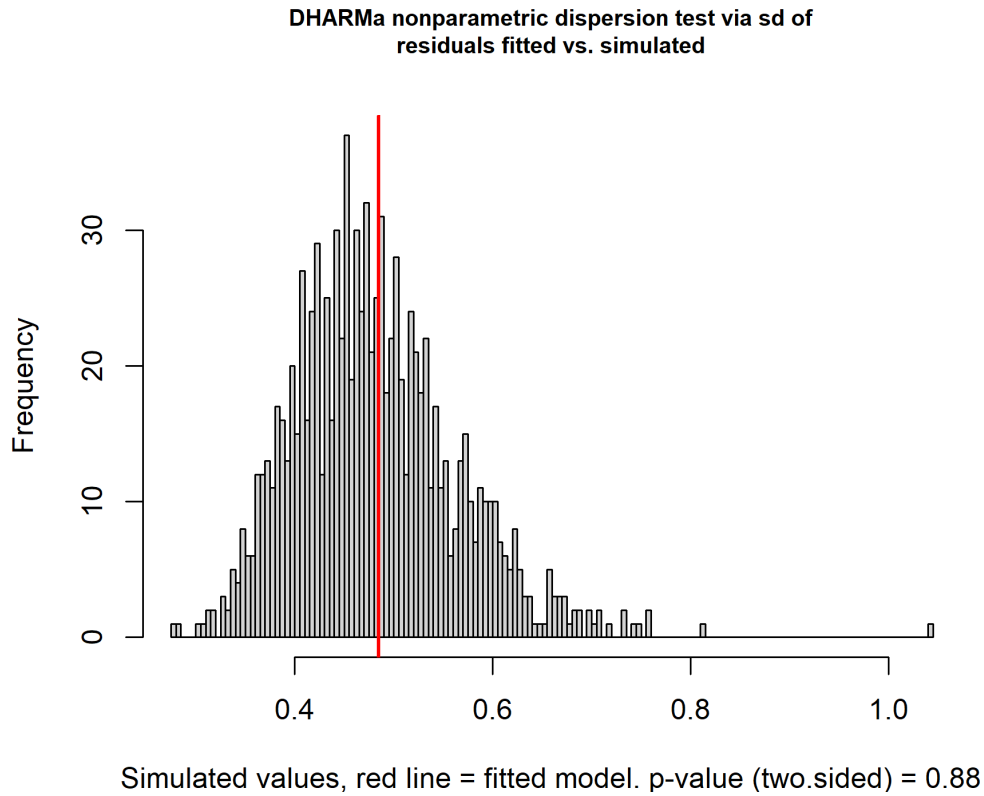


Figure 11: Check for whether the model over- or under-estimates how spread out the counts are.

The outlier test gave a less clean answer, and is worth reporting honestly rather than picking whichever result looks better. A bootstrapped test, generally the more reliable option at this sample size, found the observed outlier frequency (0.0046) not significantly different from what the fitted model would expect (0.0019, $p = .06$). The alternative binomial-based test built into DHARMa's default residual plot, which uses a different and more conservative approximation, flagged the same residuals as significant ($p < .001$). Since the two methods disagree and the more appropriate bootstrapped test does not reach significance, this is treated as inconclusive rather than firm evidence of a problem.

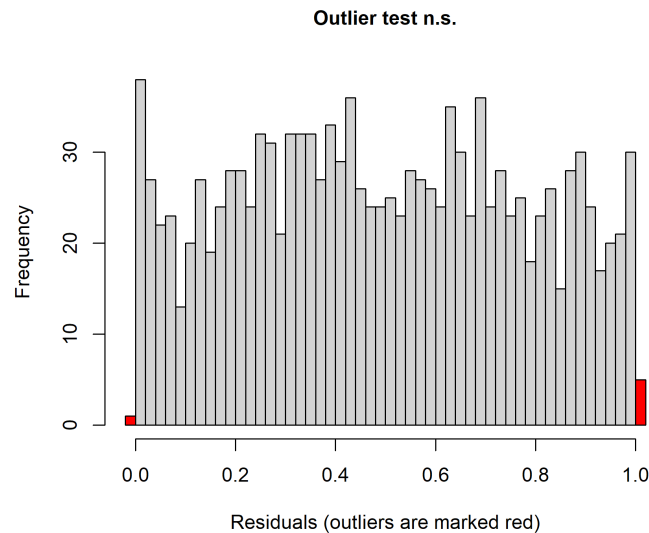


Figure 12: Check for unusually extreme strata the model struggles to explain.

The clearest issue came from the zero-inflation test (Figure 13). Comparing observed exact-zero strata against the distribution of zero counts expected under repeated simulation from the fitted model, M4 under-predicted the number of zero-count strata by a considerable margin ($\text{ratioObsSim} = 1.94$, $p < .001$): nearly twice as many strata had zero frequent attenders in the observed data as

the model’s simulations typically produced. This is a genuine limitation. Even so, M4 is retained as the primary specification reported in final result, since the two assumptions most central to a count model, residual distribution and dispersion, both held, and the substantive coefficients of interest do not depend on the model correctly reproducing the zero tail of the distribution. The zero-inflation result is carried forward and discussed as an explicit limitation in Section 6.

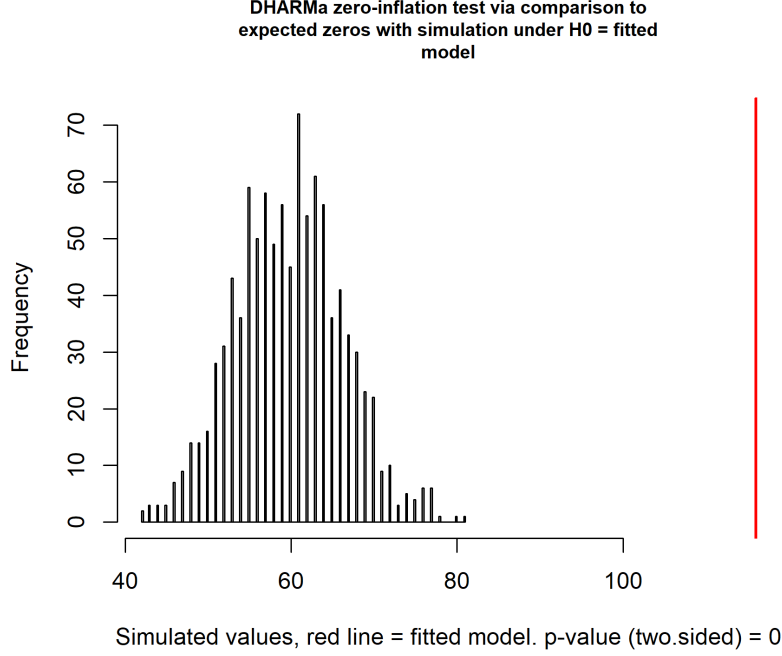


Figure 13: Check for whether the model predicts the right number of strata with no frequent attenders

A.2 Zero-inflated Negative Binomial Model

A zero-inflated model treats the observed zeros as coming from two distinct sources, where a stratum has no real chance of ever producing a frequent attendee, and the ordinary negative binomial count process already used in the primary model, which can also generate a zero by chance even where frequent attenders are possible (Green, 2021). The zero-inflated negative binomial (ZINB) extension adds a second component, a logistic regression predicting the probability that a stratum falls into the structural-zero group, alongside the negative binomial count component carried over from M4 unchanged.

To investigate the zero-inflation deficiency identified above, a zero-inflated negative binomial (ZINB) extension was fitted, with the zero-inflation component conditioned on department type, the variable most plausibly associated with structural zeros given its dominant effect on the count process (Section 3.3). This substantially reduced the zero-inflation discrepancy, with the ratio of observed to expected zero-count strata falling from 1.94 to 1.41. The residual discrepancy remained statistically significant ($p < .001$), which is expected given the large number of strata ($n = 1,307$) makes even modest departures from the fitted distribution detectable. The improvement indicates that department type accounts for a substantial share of the excess zeros, consistent with minor injury units structurally producing few or no frequent attenders in many strata, but not all of it: some residual zero-inflation likely reflects additional sparsity in strata with small population offsets (e.g. specific age $\times$ deprivation $\times$ board combinations), which a single zero-generating covariate cannot fully capture. The original negative binomial model (M4) is retained as the primary specification, since its coefficients remain directly interpretable as attendance rate ratios and the substantive conclusions do not depend on correctly reproducing the zero tail. The ZINB result is reported here as evidence narrowing the source of this limitation rather than as a change to the primary model.

We also tested extending the zero-inflation formula to include deprivation alongside department

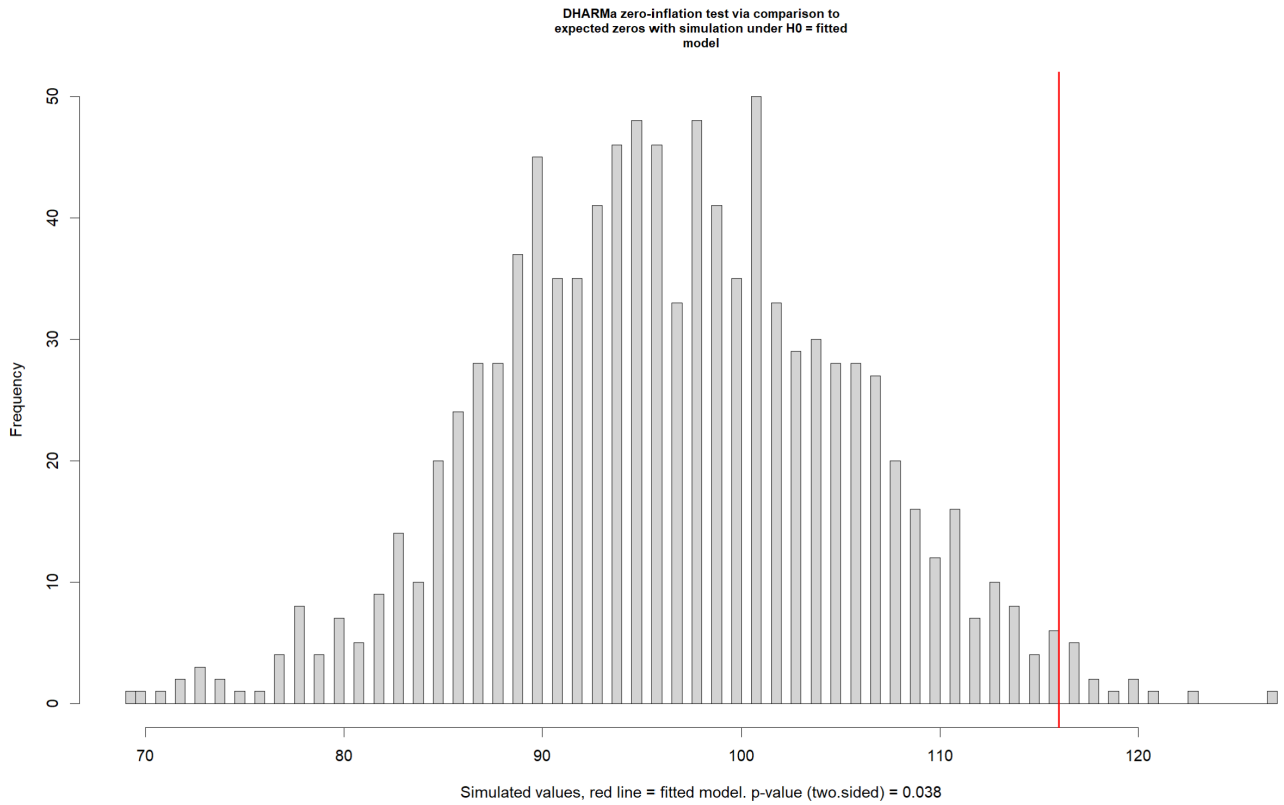


Figure 14: Zero-inflation test for the ZINB extension with department type in the zero-inflation component, showing the reduced discrepancy between observed and expected zero-count strata.

type, since deprivation is the model’s other structurally sparse predictor. This brought the zero-inflation ratio down a little further, to approximately 1.2, but it brought back a stability problem we’d run into earlier in model development. The estimates for deprivation quintile 2 and quintile 4 in the zero-inflation component came with standard errors several orders of magnitude larger than the estimates themselves ($SE = 1690$ and 2983 respectively). In practice, this means there simply aren’t enough zero-count strata at these particular deprivation levels to support a reliable estimate once department type is factored in too. Given that, we didn’t carry this version forward. The slight improvement in zero-inflation fit wasn’t worth the instability it introduced elsewhere in the model.

A.3 Interaction Model between Age and Sex

The main text reports the additive model. This section gives the full specification and output of the age-by-sex interaction model tested as a robustness check. The interaction model added `age * sex` in place of the additive `age + sex` terms, with all other covariates unchanged:

```
frequent ~ age * sex + deprivation + department_type +
  rate_heartfailure + rate_depression + offset(log(population)) + (1 | hbt)
```

A likelihood ratio test against the additive model showed a significant improvement in fit ($\chi^2 = 14.53$, $df = 5$, $p = .013$), with the interaction model’s AIC coming in at 11119.4 against 11124 for the additive model. Table 4 gives the full coefficient table.

Table 4: Age-by-sex interaction model, conditional model coefficients

Predictor	β	Rate ratio	95% CI	p
Age (ref: Under 18)				
18–24	−0.124	1.13	0.92–1.39	.239
25–39	−0.123	0.88	0.72–1.08	.233
40–64	−0.344***	0.71	0.58–0.87	<.001
65–74	−0.149	0.86	0.70–1.06	.150
75 plus	0.534***	1.71	1.40–2.08	<.001
Sex (ref: Female)				
Male	0.156	1.17	0.96–1.43	.128
Deprivation (ref: Quintile 1, most deprived)				
Quintile 2	0.068	1.07	0.94–1.22	.312
Quintile 3	0.201**	1.22	1.06–1.40	.004
Quintile 4	−0.161*	0.85	0.74–0.98	.022
Quintile 5 (least deprived)	−0.766***	0.46	0.41–0.53	<.001
Department type (ref: Type 1)				
Type 3	−2.167***	0.11	0.10–0.13	<.001
Health Board disease prevalence				
Heart failure rate	0.646*	1.91	1.12–3.26	.018
Depression rate	−0.156*	0.86	0.75–0.97	.018
Age $\times$ Sex (ref: Under 18 $\times$ Female)				
18–24 $\times$ Male	−0.444**	0.64	0.48–0.86	.003
25–39 $\times$ Male	−0.222	0.80	0.60–1.07	.127
40–64 $\times$ Male	−0.138	0.87	0.66–1.15	.335
65–74 $\times$ Male	−0.020	0.98	0.74–1.30	.890
75 plus $\times$ Male	0.038	1.04	0.78–1.38	.790

significant at *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The Health Board random effect variance was 0.095 (SD = 0.309), close to the additive model. The dispersion parameter was 2.01. Reference categories are age under 18, female, deprivation quintile 1 (most deprived), and department Type 1.

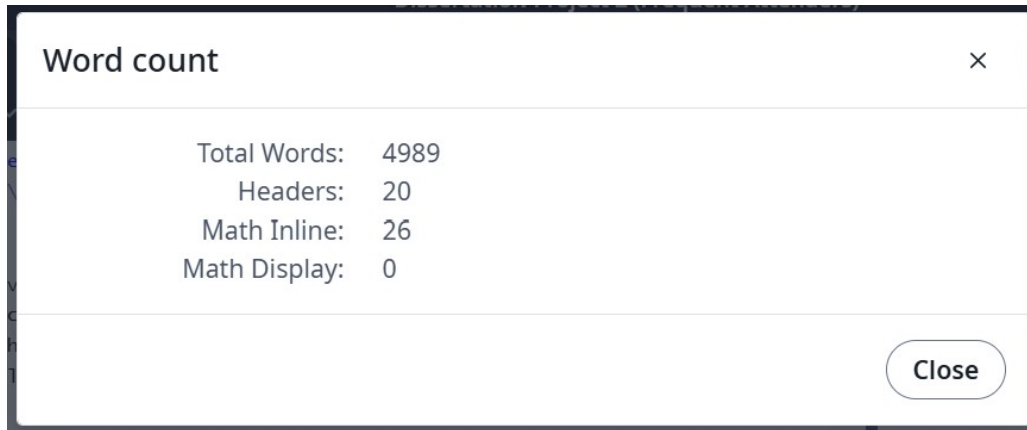
Only the youngest age band showed a significant interaction. Men aged 18–24 attended less often relative to women in the same band than the additive model predicted, while the other four age bands showed no significant departure from the additive pattern. This is a narrower result than the crossover suggested by the descriptive data, which spanned the full age range rather than concentrating in one band. The gap likely says less about age and sex working together in general and more about deprivation and department type, both dominant predictors in this model, absorbing part of the pattern once they enter the equation. Once they do, only the youngest men stand apart from what the additive model expects.

Five extra terms bought a small, narrow improvement over the additive model. That trade-off did not justify the added complexity, so the additive model stands as the reported specification. The young-men result is carried forward as a flag for further work rather than folded into the main age effect.

A.4 Code and Reproducibility

The full analysis pipeline, from data cleaning through model fitting and diagnostics, along with the reference list, is available at <https://github.com/fenandadwithak/dissertation2>.

A.5 Word Count

A screenshot of a 'Word count' window from the Overleaf editor. The window has a title bar with the text 'Word count' and a close button (an 'x' icon). The main content area displays four statistics: 'Total Words: 4989', 'Headers: 20', 'Math Inline: 26', and 'Math Display: 0'. At the bottom right of the window is a 'Close' button.

Word count		×
Total Words:	4989	
Headers:	20	
Math Inline:	26	
Math Display:	0	
		Close

Figure 15: Word Count with Overleaf